



FACULTY OF COMPUTING AND INFORMATION  
TECHNOLOGY

**FINAL YEAR PROJECT (FYP)**  
**INFORMATION FOR STUDENTS**

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## Introduction

All Bachelor of Computer Science (BCS), Bachelor of Information Systems (BIS), Bachelor of Information Technology (BIT) and Bachelor of Software Engineering degree students are required to undertake a substantial project in their final year. Each project is supervised by a lecturer (herewith referred to as the **supervisor**), and assessed via written reports and presentations.

The final year project provides students with hands-on practical experience in planning and managing a project. The students will need to select and apply relevant methodologies, theories and skills in systems development for application or research-based projects. The planning of the project requires a degree of inventiveness and knowledge of the subject areas of the programme. In addition, the final year project requires students to apply relevant fact-finding techniques, modelling tools and techniques, and the process of designing a computer system. Besides, the course also intends to equip students with skills to plan, manage, produce and evaluate the project that incorporates good system design, such as workflow design, user interface design, database design, report design, etc.

In addition, the final year project also requires the students to demonstrate leadership skills as it is challenging and demands a tremendous amount of self-discipline, self-motivation, good time management and all-out effort from the student. Students will also need to exercise their problem-solving skills, creativity, and communication (both verbal and written) skills.

The final year project (FYP) is conducted over the following two courses:

- **BACS3403 Project I** and
- **BACS3413 Project II**.

All forms, guidelines, templates, information, etc. relevant to the FYP will be posted at the FYP **Google Classroom**.

## Academic Honesty

All students are expected to pursue their scholastic careers with honesty and integrity. Therefore all projects submitted must be original and not any that has previously been submitted for assessment at the institution or elsewhere. Academic dishonesty is completely unacceptable and will not be tolerated in any form by this institution. Students are reminded that failure to maintain honesty and integrity in any academic pursuit in the University leads to serious consequences that may include being expelled from the institution.

Academic dishonesty includes, but is not limited to, cheating, plagiarism, collusion, the submission for credit of any work or material that is attributable in whole or in part to another person, taking an examination or doing project assignments for another person, copying answers of another person or letting others copy answers in examination, tutorials or project assignments, or any act designed to give an unfair advantage to a student, or the attempt to commit such acts.

All work submitted must be original and NOT previously submitted for assessment in this institution or elsewhere. A modified piece of work previously submitted by other students for assessment is also strictly not acceptable. All information obtained from other sources must be duly acknowledged. In particular, sources of information discovered on the Internet need to be cited as those from any other publications. There are many guides for academic writing that show how such sources should be cited. Therefore, in producing the final project report, the student should beware of plagiarism. Note that plagiarism could mean any of the following:

- Direct copying of phrases and/or passages, or paraphrasing another author's work without proper referencing or citation.
- Copying another person's work, in part or in whole.
- Writing your work in conjunction with other sources without prior permission.
- Submitting work that has already been submitted for other assessment by other students or seniors.

Refer to the University's **Plagiarism Policy** for details.

Students may refer to books or other material to assist them in their projects, especially in documentation. However, students are reminded that in writing up the report, they should only describe what they have **actually done** and not simply extract or copy notes representing a description of what the stages are all about. Bear in mind that the supervisor is not looking for the extract of notes about a subject matter or the stages of a project life cycle but a write-up of what **had actually been carried out** by the students.

## Research Ethics and Private Data Protection

Students who wish to carry out market survey or research survey, such as questionnaire, interview, etc. are required to pay attention to the following.

All survey questions drafted by students must be screened through by their respective supervisors, and it must be ensured that they follow TAR UC's **Ethical Guidelines for Research & Development and Professional Practice**.

- **Private Data Collection:** Ensure NO Private Data such as name, IC number, address, contact number, and so on, is collected. If private data must be collected, then prior approval from the Faculty Ethics Committee must be obtained.
- **Demographic Data Collection:** It is permissible to collect socio-demographic data such as gender, age, experience, etc. However, if the objectives of the project/research do not reflect the requirement of these demographic data as per stated in the TAR UC ethics guide, they should not be included as part of the survey.
- **Sensitive Data Collection:** Avoid the collection of sensitive data, such as data related to politics, discrimination, religious view, sexual orientation, gender bias, etc. Discrimination against any individual, either directly or indirectly on grounds of race, gender, disability, age, marital status, socio-economic background, political or religious belief is strictly prohibited.
- **Vulnerable Subjects:** Research involving vulnerable subjects, such as under-aged children, although with permission granted by their parents, must obtain approval from the Faculty Ethics Committee prior to the research.
- **Data Collection Involving School Students:** Prior approval from the Ministry of Education and Faculty Ethics Committee must be obtained.
- **Data Collection Involving TAR UC Students:** Prior approval from the Faculty Ethics Committee must be obtained.
- **Data Management:** Data collection from humans must be addressed and managed properly. The collected responses must be kept confidential and safe. It is strictly prohibited to share the collected data with other unrelated persons without consensus from the data owner. Do not publish an individual's response on Facebook, any Internet site, etc., which can be accessed by other unrelated persons. The collected data that are no longer useful can be discarded but must be done carefully to ensure no leakage would occur.

## Intellectual Property Rights

The University regards IP as knowledge and its creative application, and includes a variety of legal rights (known as 'IPR') which protects the applications of ideas and information that may be of commercial and/or academic value. IP most relevant to the University's activities include patents, copyright, design rights, trade secrets and know-how. In practical terms, all materials generated by staff and/or students should be regarded as potentially being IP and generating IPR, including students' final year projects. This policy is intended to provide guidance for the benefit of staff and/or students on the University's position concerning the key principles of ownership, protection and exploitation of IP. The University treats students and staff equally in relation to IP. This policy applies to all full-time employees, to all enrolled and/or registered students; it also applies to part-time employees (including where their contract of employment includes any non-teaching duties and/or final year or research student project supervision).

Refer to TAR UC's **Policy on Intellectual Property Rights** for details.

## FYP Process

The overall FYP process is outlined in the table below. Refer to the FYP's Google Classroom for the **deadlines** of each task/deliverable for your cohort.

| No.                | Task                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project I:</b>  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 1                  | a. Find supervisor (refer to the <a href="#">FYP Supervisor List</a> for the list of supervisors assigned to your programme)<br>b. Select project title                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 2                  | Check <a href="#">Supervisor-Student List</a> (Note: Supervisors will be assigned to the students who did not find a supervisor by the stipulated deadline).                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 3                  | Submit <a href="#">Project Proposal</a> to supervisor.<br>Note: The project proposal must be approved by both the supervisor and the moderator.                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 4                  | Submit <b>Chapter 1 Introduction</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5                  | Submit <b>Chapter 2 Literature Review</b> or <b>Chapter 2 Related Work</b><br>Application for Ethical Review (if required)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 6                  | Submit <b>Chapter 3 Methods and Requirements Analysis</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 7                  | Submit <b>Chapter 4 System Design</b> or <b>Chapter 4 Research Design</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 8                  | Submit <b>Project I Portfolio</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Project II:</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 1                  | Prepare <b>Test Plans</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 2                  | <b>System Preview</b> with Supervisor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 3                  | <b>Final System Testing</b> with Supervisor and Moderator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 4                  | Submit Draft <b>FYP Report*</b> and <b>Poster</b><br>*The FYP Report comprises the updated chapters from Project I and additional chapters such as Chapter 5 Implementation and Testing; Chapter 6 Conclusion; etc.                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5                  | Submit <b>Final FYP Report</b> ( <a href="#">Softcopy</a> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 6                  | <b>Digital Submission of FYP artifacts</b><br>To submit all the final versions of the following items as a folder (with your name and programme) <a href="#">via Google Classroom**</a> (**details to be provided at a later date):<br>a. FYP Report <b>Full Text</b> (use the template <a href="#">FYP Report Template.dotx</a> )<br>b. FYP <b>Abstract</b> (use the template <a href="#">AbstractTemplate.docx</a> )<br>c. FYP Search <b>Keywords</b> (use the template <a href="#">KeywordsTemplate.docx</a> )<br>d. <b>Poster</b> (use the template <a href="#">PosterTemplate.pptx</a> )<br>e. <b>Source code</b> |

Students are expected to show the draft version of each chapter/deliverable to the supervisor for comments before submitting the final version of each chapter/deliverable. Note that the supervisor will take the quality of the initial work submitted (i.e. the draft versions) into

consideration when awarding marks. Therefore, make all effort to ensure that your work is correct and presentable in all aspects.

Project supervisors **WILL NOT**:

- Carry out research for a student under his/her supervision.
- Write the project chapters/deliverables/report for the students.
- Reveal any marks or grades that are awarded to the project.
- Correct the grammatical and/or spelling mistakes in the report for the students.
- Provide support on technical aspects, e.g. teach a student under his/her supervision how to program using a certain programming language, how to do normalization, draw models, etc.

## Project Selection

A project should, among other things, contain the following elements:

1. **Relevance and contribution.** The project is within the area of specialization related to the student's programme of study. The outcome of the project is able to contribute to the IT practices, target market, or knowledge. Refer to the Programme Specialization for your programme's FYP:
  - [Programme specialization for REI FYP](#)
  - [Programme specialization for RIS FYP](#)
  - [Programme specialization for RIT FYP](#)
  - [Programme specialization for RSD FYP](#)
  - [Programme specialization for RSW FYP](#)
  - [Programme specialization for RST FYP](#)
  - [Programme specialization for RDS FYP](#)
2. **IT content.** The project is IT-related and has substantial amount of IT content.
3. **Technical Skill.** The project requires the students to write substantial amounts of programming codes, or use of IT technical skills with the aid of tools.
4. **Methodology.** The project allows the students to apply some kind of system development or research methodology.
5. **Practicality or innovativeness.** The project should have qualities of practicality and/or innovativeness. For research projects, the project should be meaningful. For application projects, it should either be an industrial project, an internal project or be aligned to an external competition. For entrepreneurial projects, the project should have a unique value proposition. Refer to [Project Types](#) and [Project Characteristics](#) for further details.
6. **Knowledge Expansion.** A project should allow the students the opportunity to expand their existing knowledge or to learn a new programming language, tools, techniques or other software.
7. **Scope.** The scope of the selected project should be such that it allows the student to complete it within the boundary of the student's knowledge and skills, and constraints of resources available, but should be sufficiently challenging enough to stretch their intellectual ability and knowledge.

Any project title that does not comply with the above requirements will strictly not be accepted.

## Project Types

A project may be one (or a mix) of the following *project types*:

- **Application projects** focuses on the development of a software application which includes web applications, mobile applications, and so on. Students would need to analyse the requirements, and then design, implement and test the system.
- **Research projects** addresses pertinent research questions whereby students will concentrate on a chosen problem or topic from their field of study and carry out the following: review the literature, identify the problem, design the research methodology, develop a prototype, interpret the results and report the research findings.
- **Entrepreneurial** - projects focused on developing applications for commercialization. Students will need to attend briefing/sessions organized by TAR UC's Centre for Business Incubation and Entrepreneurial Ventures (CBIEV).

The following table shows examples of the *typical*\* chapters that will be included in the FYP Report of the different project types:

|             | Academic <b>Research</b> | <b>Application</b> Development                                     | <b>Entrepreneurial</b>            |
|-------------|--------------------------|--------------------------------------------------------------------|-----------------------------------|
| <b>Ch 1</b> | Introduction             | Introduction                                                       | Introduction                      |
| <b>Ch 2</b> | Literature Review        | Background and/or Literature Review**                              | Background & Business Opportunity |
| <b>Ch 3</b> | Methodology              | Requirements Analysis / High Concept Document and Game Treatment** | Requirements Analysis             |
| <b>Ch 4</b> | Research Design          | System Design / Game Script**                                      | Prototype/System Design           |
| <b>Ch 5</b> | Results & Evaluation     | Implementation & Testing                                           | Implementation & Testing          |
| <b>Ch 6</b> | Future Work              | Deployment (if applicable)                                         | Deployment (if applicable)        |
| <b>Ch 7</b> | Conclusions              | Project Evaluation                                                 | Project Evaluation                |

\*Students may introduce other chapters as appropriate - please refer to your FYP Supervisor to confirm if there are other chapters or chapter headings to be included.

\*\*Select the appropriate chapter heading based on the nature of your project / project specialization.

## Project Characteristics

Regardless of the project type, a project may have one or more of the following characteristics:

### a. **Competition**

A project be aligned to and/or submitted for one or more appropriate external competitions (e.g. Innovate Malaysia Design Competition, Imagine Cup, etc).

### b. **Project Client**

*Application* projects and *Research* projects may have a *project client*. The project client may either be an *external client* or an *internal client*.

#### i. **Industrial Projects**

Projects with external clients are referred to as **Industrial Projects** - these projects have an external company/organization as the client and the software is specially developed for the client. The client is required to sign a Memorandum of Agreement (MoA) with TAR UC before the project can commence.

#### ii. **Internal Projects**

Projects with internal clients are referred to as **Internal Projects** and these projects include:

- **Smart Campus** projects - projects under the TAR UC Smart Campus project which aims to turn the KL Main Campus into a smart campus.
- **Cross-Faculty projects** - projects in which the students from a different faculty are the clients. For example, FAFB students have a business idea and the FOCS students' FYP is concerned with developing the required application / system for this business idea.
- **Departmental projects** - projects in which a TAR UC department (DECA, Library, etc) is the client.
- **Supervisor's titles** - these are projects that were identified by the respective supervisors.

## Project Proposal

Once the project has been agreed upon by the supervisor and student, **each student** in the team must complete and submit the following documents to the supervisor before the stipulated deadline:

- **FORM 1 Student Details,**
- **FORM 2 Project Proposal and**
- **FORM 3 Project Proposal Moderation**

For **Application** and **Entrepreneurial** projects, your proposal shall include the following details:

- Customer segments: Target customers or (for projects with a client) the background information of the company/organization.
- The main problems that your potential customers/users face.
- An evaluation of existing alternatives or (for projects with a client) the existing system.
- The proposed solution to the problem: What theory and background information (if any) underpins the project; The feasibility of the project; (for entrepreneurial projects) the unique value proposition (UVP).
- In what way is the proposed project novel? What advantages does your solution offer? In what ways may the project contribute to the industry or community.
- Project milestones; What software development methods will be utilised? What tools (and hardware, if any) will be used, so far as decided?

For **Research** projects, your proposal shall include:

- The research problem statement
- The proposed solution; What theory (literature) underpins the project (literature review);
- The research question to be addressed and the research hypotheses that you wish to test; Specify the target users or respondents.
- Methodology: What research methods will be utilised? What tools (and hardware, if any) will be used, so far as decided? How you will judge the validity and generality of your results, e.g. experiment; project milestones.
- In what ways may the project contribute to related research activities and knowledge pool. If it is an industrial project, provide the background information of the company.

The project proposal

- Needs to be approved by both the supervisor and moderator,
- Will be used as the basis for Chapter 1 in the FYP report, and
- Needs to be included in the Appendix of the final FYP report.

## Application for Change in Project

### a. Changes in Project after Approval

Generally speaking, students are not allowed to make major changes to the project specification after it has been approved. Any changes to a project which has already been moderated and approved can only be requested by the students in the **early** stages of the project life cycle and the changes requested should not affect the scope of the project significantly. Late requests for changes will not be entertained. Changes are not encouraged, as it would mean a waste of time for both the supervisor and students. In any case, students are required to seek the supervisor's prior approval for any changes requested. If the student, without prior discussion and approval by the supervisor, made any changes, the project would be rejected regardless of how much the project has progressed. Even if the supervisor ultimately approves the changes, students would still have to start the project life cycle all over again from the beginning.

### b. Change of Project Supervisor

Once a student has selected or has been assigned a final year project supervisor, no changes are allowed subsequently. However, under special circumstances, the change of supervisor may be allowed by the FYP Committee. The student will then be informed by either the former supervisor or the FYP Committee about the changes.

## Student Attendance

Students are required to attend a series of project progress meetings with the supervisor on a regular basis from the beginning till completion of the project. Although there is no fixed frequency or number of meetings with the supervisor, it should average out to at least twice a month, from the time the [Supervisor-Student List](#) is released. Constant absence from meetings will result in stern action being taken against the students, or in students losing valuable marks.

The main objectives of progress meetings are as follows:

- For students to report the progress of the project to their supervisors. Other than updates on the progress made, students may also highlight problems faced, solutions being considered, project variance, etc.
- For students to consult their supervisors to discuss about problems encountered during the project, obtain clarification on their tasks, present their work, etc.
- For supervisors to assess students' performance of the projects.

## Deliverables and Assessment - General Instructions

Each student is required to submit a written **FYP Report** and other artifacts at the end of your project. You will progressively write and submit the FYP report chapters and other deliverables to your supervisor throughout BACS3403 Project I and BACS3413 Project II. Refer to the [FYP Process](#) for an overview of the submissions.

Download the **FYP Report Template** from the Google Classroom. You may make any appropriate amendments to the structure of the FYP report.

You should submit a *draft* version of each deliverable to your supervisor for comments during your project meetings. Note that

- You should submit the drafts very much ahead of time in order to allow your supervisor to have enough time to review them and for you to make the necessary improvements;
- Your supervisor is only expected to read one draft of each deliverable. Hence, for your benefit, the draft should be as complete as possible so that you get more complete feedback;
- In awarding marks for the deliverables, the quality of the initial work will also be taken into consideration. Therefore, you should invest sufficient time and effort in carrying out all project work.
- The draft versions of deliverables can be submitted either in soft copy or hard copy depending on the prior arrangement with the supervisor. The supervisor would then provide feedbacks on changes or amendments, if any, with regards to the report.

In order to ensure fairness and consistency in the assessment of students' projects, selected samples of student projects will be subjected to moderation by another supervisor who would act as a *moderator*.

## BACS3403 Project I Deliverables

All diagrams, draft work, deliverables, and other project related documents must be properly organised and maintained in a project file. Students must supply and maintain their own files. The project file will be used to hold all documentation (whether drafts or final) pertaining to the project. Where the file is kept by the students, it must be brought along for EVERY project progress meeting. This is necessary as the supervisors would invariably make references to previous copies of reports for written comments, amendments, past discussions and suggestions made. In addition, this would also allow the supervisors to assess the students' project management skills and initiatives demonstrated.

The Project I deliverables shown below are to be submitted progressively by each student as individual work (refer to the FYP deadlines for your cohort for each deliverable submission deadline):

- a. Project Proposal ([Form 1](#), [Form 2](#) and [Form 3](#))
- b. Chapter 1 Introduction
- c. Chapter 2 Literature Review
- d. Chapter 3 Methods and Requirements Analysis
- e. Chapter 4 System Design
- f. Project I Portfolio

At the end of the semester before the given deadline, the **project portfolio** consisting of all the project deliverables (i.e. items a, b, c, d and e - can be in either softcopy or hardcopy, based on the supervisor's requirement) must be submitted to the supervisor for assessment purpose.

## BACS3403 Project I Assessment Rubrics

Refer to the [Project I Assessment Rubric](#).

## Progression to BACS3413 Project II

After having completed the requirements for BACS3403 Project I, the students are required to continue developing a working system or prototype in BACS3413 Project II, following their system planning and designs in BACS3403 Project I. To allow the students to proceed to BACS3413 Project II, the students must obtain a PASS for BACS3403 Project I. In the case where a student does not obtain a PASS in this course, he or she is required to **repeat** BACS3403 Project I in the subsequent semester that offers the same course.

## Repeat BACS3403 Project I

Any student who does not meet the minimum threshold, i.e. 50% of the overall assessment, is deemed to have failed the course.

- Any student who does not pass either BACS3403 Project I or BACS3413 Project II shall then register for the failed course during the next offer (the student should check the available semester when the course will be offered again on the FOCS notice board). The student must contact the FYP Committee or respective Programme Leader to inform about his/her plan on repeating the course. The student will then be assigned a project supervisor by the FYP Committee.
- When repeating the course, the student is expected to complete the course alone.
- Students repeating BACS3403 Project I may propose a new project title and scope.

## BACS3413 Project II Deliverables

The deliverables for Project II are as follows:

### I. Formal Project Presentation

Each student will be assessed on his/her ability to present the project to the supervisor and moderator either via a 5-minute

- **Formal presentation.** In your presentation, you should give an overview of your project and describe what you have achieved. Your task is to convince them that your project work is interesting and of high quality.

or

- **Pitch.** This is for entrepreneurial projects. Your task is to convince them that your project idea is worth investing in.

### II. Final System Testing

For the final system testing, selected projects will be tested by the assigned moderator. The system testing by the moderator may either be carried out simultaneously with the supervisor or separately if it is not possible to arrange a common time. In this regard, the project supervisors concerned will make their own arrangements with the corresponding moderators. However, the project testing by the moderator will usually only be carried out **once**.

For projects with clients, the moderator for the final system testing will be the project's **client** and the final system testing will normally be considered as the **user acceptance test (UAT)**.

As part of the preparations for the final system testing, students are required to

- Ensure that there is sufficient data in the database tables (and in the other persistent forms of data storage) to demonstrate the working functionalities and features of their system;
- Prepare sample data and bring their test cases on the day of testing and to use them to demonstrate the system to the supervisor and moderator;
- Print the requirements document for the moderator's reference;
- Demonstrate understanding of the work that they undertook.

The final system testing must be preceded by either a 5-minute formal project presentation or pitching by each student.

Important:

The final system testing is a crucial part of the FYP assessment. A student must **pass** the system testing before he/she can be given a pass in the FYP.

### III. Poster

Each project team should produce ONE (1) poster with the following requirements:

- Format: Powerpoint

- Size: **A1** size
- Submission mode: Softcopy

Download and use the appropriate [Poster Template](#) from the Google Classroom.

#### IV. FYP Report

**Each** student must write and submit a complete FYP Report individually. The FYP Report is a very important component of the FYP. Each student is responsible to compile his/her FYP Report **separately** and **individually**, even though he/she may have worked as a member of a project team in a joint project. The FYP Report will not only be assessed by the supervisor, but possibly by a project moderator and/or an external examiner. For details on the format of the FYP Report, refer to the [FYP Report Format](#). The **FYP Report Template** is available at the Google Classroom.

Each student only needs to submit the FYP report in softcopy. The FYP reports with Grade B and above will be uploaded to the TARUC Institutional Repository (IR). In addition, each student must also provide a digital submission of the required project artifacts, as will be instructed in the [Guidelines for Digital Submission](#).

#### BACS3413 Project II Assessment Rubrics

Refer to the Google Classroom for the [Project II Assessment Rubric](#).

## FYP Report Format

The **FYP Report Template** is provided for your reference and guide. You may change the table of content, any chapter title and its content where appropriate to suit the thesis writing relevant to your project. You may include more chapters for more detailed explanations. However, do focus on writing a quality thesis, rather than on merely producing a massive number of pages. The writing of an extended and scholarly dissertation is assessed based on the mettle of the student on how much is your grasp of the subject. A good thesis should be clear and concise, and presenting its authenticity, genuineness, quality of research, methodology, analysis and discussion.

You may download the **FYP Report Template** from the Google Classroom.

Students are required to comply with the following format of the FYP report to be submitted.

- **Front Cover.** The front cover should be black with gold-coated text.
- **Back Cover.** The back cover is black in color.
- **Line Spacing.** Typed material should be 1.5 line-spaced.
- **Font Size.** A body text of font size 11 or 12 is required while for headings and sub-headings a larger font size must be used.
- **Font Style.** Use New Times Romans for body text. Main headings and subheadings should be clearly worded and in suitable font styles (e.g. Arial).
- **Headers and Footers.** Appropriate footers and headers should be used to enhance clarity and appearance.
- **Margin.** All pages should have a margin of an adequate width of 3 cm (1 inch) all around.
- **Page numbering.** Ensure that all pages (except cover page) are numbered.
- **Paper Size.** Use A4 paper, single-sided printing.
- **Number of Copies.** Only **ONE SOFTCOPY** of the project report needs to be submitted to the supervisor by **each student** in the team.

### Word or Page Limit

Although there is no word or page limit in the final project documentation, the supervisor will be looking for quality of work done rather than the quantity. Very commonly, students are obsessed with the thickness of the documentation. Wasted efforts are often invested to make the documentation as thick as possible, for example, by copying notes from books, etc. The thickness of a project report has no meaning if the qualitative aspects have not been properly taken care of.

### Language and Writing Style

The document should be formal in style and should be written using **passive voice**. Some examples are given below:

| Active voice                    | Passive voice (recommended)                     |
|---------------------------------|-------------------------------------------------|
| I carried out an interview ...  | An interview was carried out ...                |
| In my opinion, I think that ... | Based on the facts collected ..., it can be ... |

As far as possible, personal pronouns, (e.g. I, We, You etc) should only be used sparingly and only should be restricted to areas needed to be exerted, emphasised or highlighted in a personal way, for example, under areas of 'personal reflections'. As the project documentation describes various stages which the students have undertaken, it is important that these different stages are written in proper sequence with clear flow of ideas and logically developed.

### Proofreading and Editing

Before submitting the project documentation students have to ensure the document has been thoroughly checked and edited for spelling errors, grammatical correctness, proper sentence structure, logical arrangement and flow of ideas. This aspect is important as it reflects the quality of work submitted by the student and hence, does influence the marks being awarded. Students are strongly encouraged to make use of a grammar checker and / or to get help from an English mentor where available, or any other means to ensure that the written documentation is free from the above errors.

In addition, also ensure that all figures and tables are correctly numbered and titled. Each figure/table must be explicitly referenced from the paragraph immediately before or after the figure/table. Each appendix should also be correctly numbered and titled. There should also be a reference to each appendix from one or more chapters of the FYP report.

## Repeat BACS3413 Project II

Any student who does not meet the minimum threshold, i.e. 50% of the overall assessment, is deemed to have failed the course. S/he shall then register for BACS3413 Project II during the next offer (the student should check the available semester when the course will be offered again on the FOCS notice board). The student must contact the FYP Committee or respective Programme Leader to inform of his/her plan on repeating the course. The student will be assigned a project supervisor by the FYP Committee.

Students repeating BACS3413 Project II are expected to continue developing a working system or prototype system within a given timeframe following the functional and non-functional requirements, system planning and design as done in BACS3403 Project I. The student is expected not to make major changes of the project scope as planned. Any change of project scope must be discussed with and approved by the supervisor.

## Programme Specializations

### A. Programme Specialization for REI FYP

The course provides opportunity to students to apply their technical abilities and knowledge in enterprise information systems. This includes development, research or case studies in enterprise information systems architecture, development and implementation of enterprise information systems, advanced technology to improve business information systems, business processes and workflow, enterprise resource planning (ERP), decision support systems, business intelligence, e-commerce and system administration. Besides, opportunities to work with industry through real-life projects on software development, implementation of enterprise information systems and data analysis would be given.

Remark: Although the students are also allowed to develop Web or mobile application, but the aim of the application is to bring advantages to the enterprise/business organization. The students can also focus on the business processes as well as research into business intelligence area.

The selected project title should be one that is software engineering (SE) related and includes a substantial amount of SE content. Examples of projects include, but not limited to the following:

- Business information systems
- Enterprise resource planning (ERP)
- Decision support systems
- Electronic systems such as e-commerce system, e-banking systems
- To improve business processes such as administration system, reporting system, scheduling system
- Web or mobile application for enterprise/business organization
- Business intelligence
- Emerging enterprise information technology such as big data

## B. Programme Specialization for RIS FYP

The course provides opportunity to students to apply their technical knowledge and skills in the information security area. This includes project development and research of computer security, network and communication security, operating systems security, Web security, and secures software/Web application development. Besides, opportunities to work with industry through real-life projects on secure software and Web application development would be given.

The selected project title should be one that is computer and internet security (IT security) related and includes a substantial amount of IT security content. Examples of projects include, but not limited to those shown in Table 1.

**Table 1: Examples of Projects**

| Application Project                                                                                                                                                                                                                                                                                                                                                                                            | Research Project                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Develop security portal for online system</li> <li>• Develop mobile security features</li> <li>• Develop information security system</li> <li>• Develop networking security system</li> <li>• Develop Biometric password enhancement</li> <li>• Develop Malware/Virus protection shell</li> <li>• Develop security library plug-in to existing application</li> </ul> | <ul style="list-style-type: none"> <li>• Encryption algorithms</li> <li>• Malware and viruses</li> <li>• Networking security</li> <li>• Mobile devices Security</li> <li>• Database server security</li> <li>• Database application security</li> <li>• Digital information privacy</li> <li>• Security and Privacy protection in Social Engineering</li> <li>• Online system security</li> <li>• Information system security</li> <li>• Research into Security management (project management based)</li> <li>• penetration testing enhancement</li> <li>• Ethical Hacking and Prevention</li> <li>• Biometric authentication</li> <li>• System vulnerability</li> <li>• Cloud computing security</li> </ul> |

## C. Programme Specialization for RIT FYP

This course provides opportunity to students to apply their technical knowledge and skills in the sectors of Web development, mobile computing and networked computer systems. These include project development of website & Web applications, mobile applications, Web services, and Web-based/networked information systems. Research can be conducted in the areas of emerging Web engineering and technologies, advanced computer network/Internet technology and security, and advanced mobile technologies. Besides, opportunities to work with industry through real-life projects on Web application and mobile application development would be given.

Remark: The students are allowed to do research on the 3 main areas: Web, mobile or networked system, to produce projects with substantial amount of innovation (or invention of new ideas), which may not necessarily be much practical to the current industries. This group of students should **NOT** be allowed to produce any standalone computer desktop application.

The selected project title should be one that is internet technology (IT) related and includes a substantial amount of IT content. Examples of projects include, but not limited to the following:

- Develop hybrid web applications that are cross-platform development.
- Develop a web or mobile application utilizing Media Capturing API, Video Streaming API, Web Audio API, Device API and etc.
- Improving trust management for Web Applications
- Web or Mobile Accessibility for the disabled
- Any mobile native application and mobile web application for smart devices
- Any Cloud computing application
- Grid computing
- Client-server networked system
- Peer-to-peer system
- Emerging Web technology such as Web 3.0
- Electronic system such as e-commerce, e-banking, e-customer service

## D. Programme Specialization for RSD FYP

This course provides opportunity to students to apply their technical knowledge and advanced programming skills in software & information systems development, network communications and information management. This includes project development and implementation of industry-relevant information systems, distributed computer systems, Web services, communication technology, communication-mediated applications, and software, mobile & Web applications for commercial purpose. Besides, opportunities to work with industry through real-life projects on software development and implementation of information systems would be given.

Remark: Unlike Software Engineering projects, the software/applications to be developed are more practical and industrial-based rather than being proof of concept. The students are allowed to build any types of software or information systems, but the system should be complete for immediate industrial use.

The selected project title should be one that is information system development (ISD) related and includes a substantial amount of ISD content. Examples of projects include, but not limited those shown in Table 1:

**Table 1: Examples of Projects**

| Research Project                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Application Project                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• To apply the emerging techniques to improve ERP systems. For example, the evaluation of the success of a Scrum-based ERP implementation method</li> <li>• To study the impact of Information Quality on strategic decision-making</li> <li>• To apply emerging technologies in facilitating information system. For example, Using RFID Technology in Outpatient Logistics: An Analysis of Its Potential and Acceptance</li> <li>• To automate process within a workflow management system. For example, a research towards bridging the gap between Information System Architecture and Business Architecture based on architecture patterns</li> <li>• To develop a new model in information system development. For example, to develop E-business based on a situational maturity model</li> <li>• To develop and validate a methodological framework to support Business Process Redesign initiatives, for example in the healthcare domain</li> <li>• To recommend and set up a quality database for analysis of software development process information</li> </ul> | <ul style="list-style-type: none"> <li>• Budget approval systems projects</li> <li>• Office management systems</li> <li>• Sports management systems</li> <li>• Railway reservation systems</li> <li>• Cashew industry management systems</li> <li>• Payroll management systems</li> <li>• Crime information management systems</li> <li>• Mobile store management systems</li> <li>• Electronic Shop Management software</li> <li>• Hostel Management Systems using Oracle</li> <li>• Dental Management systems</li> <li>• E-School Management Systems</li> <li>• Time Attendance Systems</li> <li>• Point of Sales system</li> <li>• Accounting information systems for SME in Malaysia</li> </ul> |

## E. Programme Specialization for RSW FYP

This course provides opportunity to students to apply technical knowledge and programming skills in the area of software engineering. Projects to be developed includes application software such as education, communication, computer-aided system, data management, desktop widgets, text/media editing, government, information science, simulation, transport, multimedia, business, etc; and system software such as utility software and operating system. Research can be conducted to study the advanced software technology, software design, the life cycles of software engineering or software management to maintain high-quality software in a systematic, controlled and efficient manner, human computer interaction and to design intelligent software/applications. Besides, opportunities to work with industry through real-life projects on software development would be given.

The selected project title should be one that is software engineering (SE) related and includes a substantial amount of SE content. Examples of projects include, but not limited to:

- artificial intelligence
- path searching
- natural language processing
- fuzzy logic
- neural network
- image & video processing
- machine learning approach to software requirements prioritization
- software architecture optimization
- steganography (hiding private or sensitive file within an image life)
- New API for mobile apps in education, communication, computer-aided system, data management, text/media editing, transport, multimedia, business, etc
- New library or utility program in server operating system (for instance, virtualization)
- New testing and validation tool (script writing) for specific application program

## F. Programme Specialization for RST FYP

The course provides opportunity to students to apply knowledge and advanced programming skills in interactive software development, in the areas such as computer games, edutainment-based applications computer simulators, and interactive applications for training, business and mass media. This also includes research in virtual reality, human computer interaction, graphic programming and algorithm, computer animation, computer games design, advanced computer games technology, and 3D games development/design. Besides, opportunities to work with industry through real-life projects on interactive software development would be given.

The selected project title should be one that is interactive software technology (IST) related and includes a substantial amount of IST content. Examples of projects include, but not limited to those shown in Table 1.

**Table 1: Examples of Projects**

| Programming or Research Type                                                                                                                                                                                                                                                                                                                                          | Game Design type                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Develop a Game Engine using a multimedia API (Programming)</li> <li>• Utilizing Neural Networks for Pathfinding (Research)</li> <li>• Improving Shader Performance (Research)</li> <li>• Develop a VR input system using motion detectors</li> <li>• Develop a custom visualization system using a multimedia API</li> </ul> | <ul style="list-style-type: none"> <li>• Develop a game using an engine</li> <li>• Design a complete level using a modding tool with own assets</li> <li>• Develop a virtual world</li> <li>• Develop a visualization system using existing engines or tools</li> </ul> |

## G. Programme Specialization for RDS FYP

This course provides opportunity to students to apply technical knowledge and programming skills in the area of data science. Projects to be developed includes application software such as data visualisation and exploratory tool in business, health care, information science, simulation, transport, manufacturing, etc; model building such as time-series model for forecasting or machine learning model for intelligent system; and backend system to harvest big data, focusing on creating interfaces and mechanisms for the flow and access of information.

Research can be conducted to study and evaluate the advanced technology in data analytic. Besides, opportunities to work with industry through real-life projects on big data and machine learning would be given.

The selected project title should be one that is data science (DS) related and includes a substantial amount of DS content. Examples of projects include, but not limited to:

- Feature engineering
- Data mining
- Artificial intelligence, machine learning
- Natural language processing
- Anomaly detection
- Social network analysis
- Image & video processing, computer vision
- New API for machine learning approach in business, information science, simulation, transport, manufacturing, etc